VIETNAM GENERAL CONFEDERATION OF LABOUR
TON DUC THANG UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY



HOANG DINH QUY VU – 252805008

Causal Inference and Deep Graphical Representations for Educational Policy Intervention

A Probabilistic Graphical Model Study on UCI and Kaggle Benchmarks

COURSE RESEARCH REPORT

PROBABILISTIC GRAPHICAL MODELS (IT701100)
COMPUTER SCIENCE

Instructor

Nguyen Chi Thien, Ph.D.

HO CHI MINH CITY, 2026

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ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Dr. Nguyen Chi Thien for his invaluable lectures, rigorous guidance, and dedicated supervision throughout the course *Probabilistic Graphical Models (IT701100)*. I deeply appreciate his academic passion, insightful feedback on causal inference methodology, and the high standards he set for our empirical research.

His thorough instruction in Bayesian network theory, Judea Pearl's structural causal models, continuous structure learning, and interventional *do*-calculus significantly deepened my understanding of graphical models and their modern applications in causal machine learning. Under his mentorship, I acquired the skills to design controlled counterfactual experiments, implement rigorous refutation audits, and analyze complex observational tabular data with academic integrity.

I am also grateful to the Faculty of Information Technology, Ton Duc Thang University, for providing an excellent academic environment and modern computational infrastructure that supported this research.

With heartfelt appreciation, I wish Dr. Nguyen Chi Thien good health, happiness, and continued success in his research and teaching endeavors.

Ho Chi Minh City, November 2026
Author

(Sign and print full name)
Hoang Dinh Quy Vu

DECLARATION

I hereby declare that this course research monograph on *Probabilistic Graphical Models (IT701100)* was carried out independently by myself under the academic guidance and supervision of Dr. Nguyen Chi Thien; and that the empirical work and results contained in it are original and have not been submitted previously for any other degree or course credit. All experimental implementations, data splits, graphical parameterizations, and counterfactual evaluations presented in this report represent my own work.

All external datasets, algorithms, and cited literature—specifically the UCI Machine Learning Repository ID 697 dataset, the Kaggle Playground S4E6 benchmark, and the foundational methodologies by Judea Pearl, Donald Rubin, Xun Zheng, and Amit Sharma—have been duly acknowledged and explicitly cited in the bibliography.

I take full responsibility for the academic integrity and validity of the research presented in this report.

Ho Chi Minh City, November 2026
Author

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ABSTRACT

CAUSAL INFERENCE AND DEEP GRAPHICAL REPRESENTATIONS FOR EDUCATIONAL POLICY INTERVENTION

Estimating the true causal impact of educational interventions from observational tabular records is a fundamental problem in modern institutional policy design. While traditional Causal Bayesian Networks (CBNs) provide rigorous identification guarantees and transparent probabilistic reasoning through Judea Pearl's do-calculus, discrete conditional probability tables (CPDs) suffer from statistical sparsity and struggle to capture continuous or high-order nonlinear interactions. Conversely, purely correlational deep learning and machine learning algorithms achieve high predictive accuracy but fail to identify counterfactual outcomes due to confounding bias and collider stratification.

In this study, we propose and rigorously benchmark a controlled hybrid framework—**Hybrid Neural-Causal Probabilistic Graphical Models (Hybrid PGM)**—that bridges structural causal discovery, Bayesian graphical modeling, and deep neural parameterization. Focusing on the public educational success benchmark (UCI Machine Learning Repository ID 697, $N = 4,424$, binary cohort $N = 3,630$) and its large-scale counterpart (Kaggle Playground Series S4E6, $N = 76,518$), we formalize a 7-node Primary Domain Directed Acyclic Graph (DAG) with 11 directed edges, separating 24 pre-enrollment baseline covariates from 12 post-enrollment academic progression variables. We design a 2×2 factorial evaluation comparing four paradigms: baseline tabular CBN with BDeu smoothing (B_0), fixed-DAG hybrid neural outcome parameterization via PyTorch Multi-Layer Perceptrons (H_1), continuous optimization-based causal discovery via NOTEARS with temporal forbidden masking (H_2), and fully neural graphical discovery-parameterization (H_3).

Across five paired random seeds under a 70/15/15 train/validation/test split, our empirical results establish three central findings: (1) Granting scholarships ($T = 1$) yields a statistically significant, positive causal average treatment effect on degree graduation ($ATE = +22.3\% \sim +23.7\%$, $p < 10^{-15}$), confirmed by DoWhy placebo and data-subset refutations. (2) Neural CPD parameterization (H_1, H_3) markedly improves predictive probability calibration (lowering Brier score from 0.1719 to 0.1667) and discriminatory ranking (AUROC = 0.8053), while preserving structural causal invariance ($ATE \approx +0.23$). (3) Causal mediation analysis demonstrates that 78.5% of the scholarship effect operates via direct socioeconomic relief ($NDE = +16.96\%$), while 21.5% is mediated through first-semester academic momentum ($NIE = +4.65\%$). Furthermore, Cinelli-Hazlett sensitivity analysis confirms robust unconfoundedness ($RV_{q=1} = 19.09\%$), and budget-constrained counterfactual policy learning reveals that prioritizing scholarship distribution based on Conditional Average Treatment Effects (CATE) produces +164 additional university graduates annually at zero incremental fiscal cost. Cross-dataset testing on 61,578 synthetic students validates that the learned graphical causal mechanisms remain invariant under

distribution shifts (ATE = +0.2143, AUROC = 0.8537).

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LIST OF ABBREVIATIONS

AIPW	Augmented Inverse Probability Weighting (Doubly Robust)
ATE	Average Treatment Effect
AUROC	Area Under the Receiver Operating Characteristic Curve
BDeu	Bayesian Dirichlet equivalent uniform prior
BN	Bayesian Network
CATE	Conditional Average Treatment Effect
CBN	Causal Bayesian Network
CI	Confidence Interval
CPD	Conditional Probability Distribution
DAG	Directed Acyclic Graph
DL	Deep Learning
DoWhy	End-to-End Library for Causal Inference
ESS	Equivalent Sample Size
GCM	Graphical Causal Model
IPW	Inverse Probability Weighting
MLE	Maximum Likelihood Estimation
MLP	Multi-Layer Perceptron
NDE	Natural Direct Effect
NIE	Natural Indirect Effect
NOTEARS	Non-combinatorial Optimization for Tabular Estimations of Acyclic Relations
OOD	Out-of-Distribution
PGM	Probabilistic Graphical Model
RCM	Rubin Causal Model (Potential Outcomes Framework)
RCT	Randomized Controlled Trial
ROC	Receiver Operating Characteristic
RV	Robustness Value (Cinelli-Hazlett Sensitivity Metric)
SCM	Structural Causal Model

TE	Total Effect
UCI	University of California, Irvine Machine Learning Repository

CHAPTER 1: INTRODUCTION

1.1 Reason for Choosing the Topic

Student attrition in higher education represents a major socioeconomic challenge that impairs institutional sustainability, strains public educational budgets, and disrupts individual socioeconomic mobility (Martins et al., 2021). Consequently, universities and government ministries allocate substantial fiscal capital toward student financial aid and scholarship programs. However, a foundational policy dilemma persists: *Does awarding an operational scholarship ($T = 1$) causally increase a student's degree completion probability ($Y = 1$), or does the observed empirical correlation merely reflect pre-existing socioeconomic self-selection?*

Standard machine learning models (e.g., Gradient Boosted Trees, Deep Neural Classifiers) estimate observational conditional probabilities $P(Y = 1 \mid T = t, \mathbf{X} = \mathbf{x})$. As proven by Pearl (2009) and Hern'an and Robins (2020), association does not imply causation:

$$P(Y = 1 \mid T = t) \neq P(Y = 1 \mid do(T = t)). \tag{1.1}$$

In university administrative registries, scholarship recipients systematically differ from non-recipients: they typically exhibit higher secondary school entrance grades (Z_2), greater family educational capital (Z_1), and better initial solvency (Z_3). Relying on associational correlations to guide policy risks severe misallocation of scarce public capital to students who would have graduated regardless of institutional aid.

Probabilistic Graphical Models (PGMs)—specifically Causal Bayesian Networks (CBNs) and Structural Causal Models (SCMs)—provide a formal graph-theoretic framework for causal identification (Ankan & Panda, 2015; Pearl, 2009). Through Pearl's *do*-calculus and graph surgery, incoming backdoor confounding arrows into T are severed, identifying the interventional distribution:

$$P(Y = 1 \mid do(T = t)) = \sum_{\mathbf{z}} P(Y = 1 \mid T = t, \mathbf{Z} = \mathbf{z}) P(\mathbf{Z} = \mathbf{z}). \tag{1.2}$$

However, canonical discrete Bayesian networks rely on tabular Conditional Probability Tables (CPTs), which suffer from combinatorial parameter explosion and sparse-cell failures when conditioning on multi-dimensional continuous covariates (Koller & Friedman, 2009). Conversely, deep neural networks provide flexible function approximation but lack causal semantics, frequently exploiting spurious correlations and collider biases (Sharma & Kiciman, 2020; Shi et al., 2019).

Developing a **Hybrid Neural-Causal Graphical Architecture** that integrates the structural identification of graphical models with the continuous density estimation of deep learning addresses this critical scientific challenge. This research establishes an end-to-end framework to isolate true causal effects, execute adversarial refutations, uncover mediation pathways, and optimize policy allocations under real-world fiscal constraints.

1.2 Research Objectives

1.2.1 General Objective

The primary objective of this research is to develop, evaluate, and benchmark a principled **Hybrid Neural-Causal Probabilistic Graphical Modeling Framework** for educational policy intervention. By unifying Bayesian network structural identification with continuous neural CPD parameterization, this study estimates the true causal treatment effect of scholarship allocation on degree completion and derives optimal, budget-constrained institutional aid policies.

1.2.2 Specific Objectives

To achieve the general objective, this research undertakes four specific objectives:

1. **SO1 (Structural Formulation and Identification):** Formalize a 7-node domain-grounded Causal Directed Acyclic Graph ($\mathcal{G}_{\text{domain}}$) and continuous structure learning pipeline (NOTEARS with Temporal Forbidden Masking) that provably satisfies backdoor admissibility for educational intervention T.

2. **SO2 (Factorial Architecture Benchmarking):** Implement a controlled 2×2 factorial model matrix (B_0: Tabular+Domain, H_1: Neural+Domain, H_2: Tabular+Learned, H_3: Neural+Learned) to empirically isolate the distinct contributions of graph topology versus parameterization across 5-fold cross-validation.

3. **SO3 (Adversarial Causal Robustness and Mediation):** Audit causal effect stability against placebo treatment permutations, random subset refutations, and Cinelli-Hazlett unobserved confounding bounds, while decomposing total effects into direct financial relief (NDE) and academic momentum (NIE).

4. **SO4 (Policy Optimization and Out-of-Distribution Transfer):** Solve the budget-constrained policy allocation problem to maximize cohort degree completions at fixed institutional expenditure, and evaluate out-of-distribution causal invariance across 76,518 instances from the Kaggle S4E6 benchmark.

1.3 Research Questions and Hypotheses

This monograph addresses one overarching question and four specific research questions and paired hypotheses:

Overarching Question (RQ0): Does financial scholarship provision (T) exert a genuine causal effect on university degree completion (Y) after adjusting for pre-enrollment confounders, through what structural mechanisms is this effect transmitted, and how can aid policies be optimized under institutional budget constraints?

1. **RQ1 (Predictive Calibration):** Does continuous neural CPD parameterization improve held-out predictive accuracy (AUROC, Macro F1) and probability calibration (Brier score) relative to discrete tabular CPTs?

- **Hypothesis H1:** Deep neural parameterization (H_1, H_3) will outperform discrete tabular baselines (B_0, H_2) in AUROC ($\Delta \geq +0.008$) and Brier calibration ($\Delta \leq -0.005$) by interpolating across sparse multi-dimensional covariate cells.

2. **RQ2 (Confounding Distortion):** What proportion of the raw observational graduation gap between scholarship recipients and non-recipients is driven by spurious confounding?

   - **Hypothesis H2:** The true causal Average Treatment Effect (ATE) will be substantially lower than the naive observational association gap, revealing an empirical confounding overestimation exceeding 25%.

3. **RQ3 (Mediation Transmission):** What fraction of the scholarship's causal effect is mediated through early academic momentum (M_1, first-semester approved credits) versus direct economic relief?

   - **Hypothesis H3:** While direct economic relief (NDE) accounts for the majority of policy impact ($\geq 70\%$), early academic credit completion (NIE) mediates a statistically significant channel ($\geq 15\%$).

4. **RQ4 (Policy Welfare Gains):** Can individual treatment effect estimation (CATE) under finite institutional budgets enhance cohort degree attainment relative to historical merit-only allocation?

   - **Hypothesis H4:** Allocating scholarships according to estimated CATE scores will yield a statistically significant graduation surge (+100 or more net additional graduates) at zero added fiscal cost.

1.4 Research Subjects and Scope

1.4.1 Research Subjects

The subjects of this research encompass two complementary domains:

1. **Methodological Subject:** Formulation, implementation, and empirical auditing of Probabilistic Graphical Models for causal inference. This includes Causal Bayesian Networks, continuous structure discovery with temporal constraints (NOTEARS), deep neural parameterization (PyTorch Neural CPDs), semi-parametric doubly robust estimation (AIPW), and Cinelli-Hazlett sensitivity algebra.

2. **Empirical Subject:** The academic retention, course completion, and graduation outcomes of undergraduate students in administrative higher education registries, specifically evaluated on the UCI ID 697 dataset ($N = 4,424$ institutional records from the Polytechnic Institute of Portalegre) and the large-scale Kaggle Playground Series 4 Episode 6 benchmark ($N = 76,518$ records).

1.4.2 Research Scope and Boundaries

The empirical and technical boundaries of this investigation are defined as follows:

- **Institutional Scope:** Undergraduate degree programs across 17 fields of study, tracking students from initial enrollment through terminal graduation or dropout.
- **Observational Policy Scope:** Analysis of state-subsidized operational scholarships (T) administered by national and university social welfare services, evaluated after controlling for all observable pre-treatment covariates.
- **Methodological Boundaries:** We focus on DAG-based causal identification under the unconfoundedness assumption ($(Y(1), Y(0)) \perp\!\!\!\perp T \mid \mathbf{Z}$), accompanied by formal sensitivity bounds for potential unmeasured confounding. Network interference and general equilibrium labor market dynamics are outside the current monograph scope.

1.4.3 Actionability Principle for Intervention Variable Selection

A foundational axiom in policy-oriented causal inference states: *"No causation without manipulation"* (Hern'an & Robins, 2020; Pearl, 2009). When formulating an interventional query $P(Y \mid do(T = t))$, the policy variable T must satisfy the **Actionability (Manipulability) Principle**:

1. **Institutional Manipulability:** The chosen variable must be physically and administratively manipulable by university decision-makers. In higher education administrative registries, demographic traits such as `Gender`, `Age at enrollment`, `Mother's qualification`, and `Nationality` represent immutable pre-existing traits. Hypothetically asserting $do(\text{Age} = 18)$ or $do(\text{Gender} = \text{Female})$ is conceptually ill-defined and administratively impossible.
2. **Distinction from Intermediate Mediators:** Similarly, academic performance metrics such as completed credits (`Curricular units 1st sem approved`) are post-treatment endogenous mediators resulting from student cognitive exertion; university administrators cannot directly assign a student's semester credits by executive fiat.
3. **Scholarship as an Actionable Policy Lever:** In contrast, awarding an operational scholarship (`Scholarship holder`, $T \in \{0, 1\}$) represents an institutional resource allocation decision fully within the jurisdiction of the university financial aid board. Administrators possess the institutional agency to award ($do(T = 1)$) or deny ($do(T = 0)$) aid. Consequently, `Scholarship holder` serves as the ideal actionable treatment variable.

1.5 Contributions of the Study

This investigation delivers six key methodological, computational, and policy contributions:

1. **Controlled 2×2 Factorial Benchmarking:** We design and execute an end-to-end framework systematically crossing graph topologies (Domain DAG vs. Learned DAG) and

parameterizations (Tabular CPT vs. Neural CPD), isolating the specific drivers of predictive and causal performance across 5-fold cross-validation.

2. **Resolution of Continuous Structure Discovery via Temporal Masking:** We show that unconstrained continuous discovery (NOTEARS) suffers from variance sortability failure (inverting temporal causality $Y \to T$). We formulate an exact Temporal Forbidden Mask matrix that restores valid causal directionality.

3. **Vectorized $O(1)$ Counterfactual Tensor Query Engine:** We develop a PyTorch-based tensor engine that executes counterfactual interventional queries across entire cohorts in milliseconds—achieving a $> 3,000\times$ speedup over iterative variable elimination.

4. **Quantification of Confounding Bias and Causal Invariance:** We demonstrate that 31.86% of the apparent raw scholarship graduation advantage is spurious selection bias (naive gap 33.9% vs. true causal $ATE = +23.09\%$). Furthermore, neural CPD parameterization preserves exact causal invariance while improving probability calibration.

5. **Causal Mediation and Subgroup Effect Heterogeneity:** We prove that 21.53% of the scholarship's effect operates through early academic momentum ($NIE = +4.65\%$), while 78.47% acts as direct financial protection ($NDE = +16.95\%$). Furthermore, non-traditional older students (> 20 years) and indebted students exhibit treatment multipliers of $1.72\times$ and $2.03\times$.

6. **Zero-Cost Policy Optimization and Out-of-Distribution Invariance:** Reallocating the historical quota of 969 scholarships via CATE ranking generates **+164** additional graduates annually (+4.52% graduation surge) at zero added fiscal cost. Finally, we demonstrate zero-shot causal transfer invariance ($|\Delta ATE| = 0.0010$) across 76,518 instances from Kaggle S4E6.

1.6 Report Structure

The remainder of this research monograph is organized as follows:

- **Chapter 2 (Theoretical Foundations and Literature Review):** Reviews Probabilistic Graphical Models, Structural Causal Models, surveys DAG structure learning algorithms (PC, FCI, Hill Climb, GES, NOTEARS) and parameter learning methods (MLE, Bayesian BDeu, EM, Neural CPDs), and reviews semi-parametric estimators and educational data mining literature.

- **Chapter 3 (Methodology and System Architecture):** Formalizes the 7-node Primary Domain DAG, establishes the four-pillar DAG validity constraints, specifies the 2×2 factorial model space (B_0, H_1, H_2, H_3), details the Deep Neural CPD architecture and Temporal Forbidden Mask, and specifies the DoWhy refutation suite.

- **Chapter 4 (Empirical Evaluation and Core Benchmark Results):** Presents Table A (held-out predictive performance), Table B (causal interventions and ATE isolating spurious confounding), and Table C (adversarial DoWhy refutations and computational runtime), accompanied by ablation diagnostics.
- **Chapter 5 (Advanced Causal Extensions and Policy Optimization):** Explores continuous GPA and retention estimands, causal mediation path decompositions (NDE vs NIE), heterogeneous treatment effects (CATE), Cinelli-Hazlett sensitivity contours ($RV_{q=1}$), budget-constrained policy optimization (+164 graduates), and large-scale Kaggle S4E6 out-of-distribution transferability.
- **Chapter 6 (Conclusions, Limitations, and Future Directions):** Synthesizes the core theoretical and empirical findings, verifies research hypotheses, articulates actionable institutional policy directives, assesses threats to validity, and outlines future research directions.

CHAPTER 2: Theoretical Foundations and Literature Review

This chapter establishes the formal mathematical foundations and reviews the relevant literature underlying this research. Section 2.1 reviews Probabilistic Graphical Models, Bayesian Network factorizations, d-separation, and the parameter explosion dilemma of discrete CPTs. Section 2.2 formalizes Judea Pearl's Structural Causal Models, the Three-Layer Causal Hierarchy, graph surgery, and the Backdoor Criterion. Section 2.3 provides a comprehensive survey of DAG structure learning algorithms (PC, FCI, Hill Climb, GES, NOTEARS) and details continuous discovery with temporal forbidden masks. Section 2.4 presents a survey of parameter learning paradigms (MLE, Bayesian BDeu, EM, Neural CPDs), semi-parametric causal estimators (AIPW, Causal Forests), and Cinelli-Hazlett sensitivity algebra. Finally, Section 2.5 surveys the educational data mining literature, critiques black-box correlational ML approaches, and delineates the specific methodological void addressed by our proposed hybrid framework.

2.1 Probabilistic Graphical Models and Bayesian Networks

Probabilistic Graphical Models (PGMs) encode joint probability distributions over multivariate random vectors by exploiting conditional independence assertions represented in directed graphs (Ankan & Panda, 2015; Pearl, 2009). Let $\mathbf{X} = \{X_1, \ldots, X_d\}$ denote a set of random variables. A Bayesian Network (BN) is defined as a pair $\mathcal{B} = (\mathcal{G}, \mathbf{P})$, where $\mathcal{G} = (\mathbf{V}, \mathbf{E})$ is a Directed Acyclic Graph (DAG) and $\mathbf{P} = \{P(X_i \mid \mathrm{Pa}_{\mathcal{G}}(X_i))\}_{i=1}^{d}$ is a collection of Conditional Probability Distributions (CPDs) conditioned on the parental set $\mathrm{Pa}_{\mathcal{G}}(X_i) = \{X_j \in \mathbf{V} : (X_j \to X_i) \in \mathbf{E}\}$.

2.1.1 Factorization and d-Separation

Under the Local Markov Assumption, each variable X_i is conditionally independent of its non-descendants given its direct parents: $X_i \perp\!\!\!\perp \mathrm{NonDescendants}(X_i) \mid \mathrm{Pa}(X_i)$. Consequently, the joint distribution factorizes recursively as:

$$P(X_1, \ldots, X_d) = \prod_{i=1}^{d} P(X_i \mid \mathrm{Pa}_{\mathcal{G}}(X_i)). \tag{2.1}$$

Equation (2.1) reduces an unconstrained joint probability mass function of size $O(2^d)$ to $O(d \cdot 2^k)$, where $k = \max_i |\mathrm{Pa}(X_i)| \ll d$.

Conditional independencies in $\mathcal{G}$ are evaluated graphically via *directed separation* (d-separation) (Pearl, 2009):

Definition 2.1 (d-separation). A trail p between V_1 and V_m is d-separated (blocked) by conditioning set $\mathbf{Z}$ if: (1) p contains a chain $V_{k-1} \to V_k \to V_{k+1}$ or fork $V_{k-1} \leftarrow V_k \to V_{k+1}$ with $V_k \in \mathbf{Z}$; or (2) p contains a collider $V_{k-1} \to V_k \leftarrow V_{k+1}$ where neither V_k nor any descendant belongs to $\mathbf{Z}$. If all trails between disjoint sets $\mathbf{X}$ and $\mathbf{Y}$ are blocked by $\mathbf{Z}$, then $(\mathbf{X} \perp_{\mathcal{G}} \mathbf{Y} \mid \mathbf{Z}) \implies \mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}$.

2.1.2 The Parameter Explosion Dilemma in Discrete CPTs

In standard discrete BNs, tabular Conditional Probability Tables (CPTs) parameterize $P(Y \mid \text{Pa}(Y))$. For discrete target Y with support $|\mathcal{Y}|$ and parents $\{Z_1, \ldots, Z_m\}$, the number of free parameters scales exponentially:

$$K_{\text{CPT}} = (|\mathcal{Y}| - 1) \prod_{j=1}^{m} |\mathcal{Z}_j|. \tag{2.2}$$

In higher education registries containing multiple continuous demographic variables, continuous features must be coarsely discretized (e.g., binning grades into 3–5 quantiles). When $m = 5$ parents are discretized into 5 bins each, a single binary node requires estimating $(2-1) \cdot 5^5 = 3,125$ conditional probability parameters. In realistic institutional cohorts ($N \approx 3,000 \sim 4,000$), many conditioning cells contain zero observations, causing Maximum Likelihood Estimation (MLE) to produce degenerate zero-frequency predictions (0/0) or fall back on uninformative uniform priors, degrading held-out calibration.

2.2 Structural Causal Models and Pearl's Causal Hierarchy

While observational Bayesian networks model $P(\mathbf{X})$, distinct DAG structures may belong to the same Markov Equivalence Class yet prescribe contradictory policy interventions (Pearl, 2009). Judea Pearl formulated Structural Causal Models (SCMs) and established the Three-Layer Causal Hierarchy (Table 2.1).

Table 2.1: Pearl's Three-Layer Causal Hierarchy and Educational Policy Equivalents.

Layer	Mathematical Form	Typical Query	Educational Domain Meaning
L_1: **Association**	$P(Y \mid T = t)$	What does observing a scholarship tell us about graduation?	Observational correlation; confounded by socioeconomic privilege.
L_2: **Intervention**	$P(Y \mid do(T = t))$	What will happen if the university mandates aid for this group?	Institutional policy effect; graph surgery cuts incoming confounding arrows.
L_3: **Counterfactuals**	$P(Y_{T=1} \mid T = 0, Y = 0)$	Would a student who dropped out have graduated had aid been given?	Individual retrospective accountability and algorithmic fairness.

2.2.1 Structural Assignments and Graph Surgery

An SCM is a 4-tuple $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U}) \rangle$, where $\mathbf{U}$ are exogenous background variables governed by $P(\mathbf{U})$, $\mathbf{V}$ are endogenous variables, and $\mathbf{F} = \{f_i\}_{i=1}^{d}$ are deterministic structural assignments:

$$V_i \leftarrow f_i(\text{Pa}_{\mathcal{G}}(V_i), U_i), \quad \forall i \in \{1, \ldots, d\}. \tag{2.3}$$

An external physical intervention setting variable T to constant t, denoted $do(T = t)$, replaces structural equation f_T with the assignment $T \leftarrow t$. Graphically, this corresponds to *graph surgery*: all directed edges entering T from its parents $\text{Pa}(T)$ are removed in the sub-graph $\mathcal{G}_{\overline{T}}$, severing all incoming confounding paths.

2.2.2 The Backdoor Criterion and Identification Formula

Definition 2.2 (Backdoor Criterion (Pearl, 2009)). A set of variables $\mathbf{Z}$ satisfies the Backdoor Criterion relative to an ordered pair of variables (T, Y) in a DAG $\mathcal{G}$ if:

1. No node in $\mathbf{Z}$ is a causal descendant of treatment T (i.e., $\mathbf{Z} \cap \text{Descendants}(T) = \emptyset$).
2. $\mathbf{Z}$ blocks every backdoor path between T and Y that enters T through an incoming arrow in $\mathcal{G}_{\underline{T}}$.

When backdoor admissibility holds, the interventional distribution is non-parametrically identified by the Backdoor Adjustment Formula:

$$P(Y = y \mid do(T = t)) = \sum_{\mathbf{z} \in \mathcal{Z}} P(Y = y \mid T = t, \mathbf{Z} = \mathbf{z}) P(\mathbf{Z} = \mathbf{z}). \tag{2.4}$$

Correspondingly, the Average Treatment Effect (ATE) of shifting treatment from control ($T = 0$) to intervention ($T = 1$) on binary outcome $Y \in \{0, 1\}$ is:

$$\begin{aligned} ATE &= \mathbb{E}[Y \mid do(T = 1)] - \mathbb{E}[Y \mid do(T = 0)] \\ &= \sum_{\mathbf{z}} \left[P(Y = 1 \mid T = 1, \mathbf{z}) - P(Y = 1 \mid T = 0, \mathbf{z}) \right] P(\mathbf{z}). \end{aligned} \tag{2.5}$$

2.3 Causal Discovery: From Combinatorial Search to Continuous Optimization

When the true data-generating causal DAG $\mathcal{G}^*$ is unknown a priori, causal discovery algorithms infer directed graph topologies directly from observational data $\mathcal{D} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)}\}$ (Zheng et al., 2018). In modern computational libraries (e.g., `pgmpy`, `causal-learn`), structure learning algorithms fall into three fundamental paradigms:

2.3.1 Taxonomy of Structure Learning Paradigms

1. **Constraint-Based Methods (PC, FCI):** The Peter-Clark (PC) algorithm operates under the Causal Markov and Faithfulness assumptions. It initializes a fully connected undirected skeleton and executes conditional independence tests ($X \perp\!\!\!\perp Y \mid \mathbf{S}$) of increasing order (e.g., Fisher-Z or χ^2 tests) to remove spurious edges. Subsequently, unshielded colliders ($X \rightarrow Z \leftarrow Y$) are oriented, and Meek rules propagate edge directions. The Fast Causal Inference (FCI) algorithm extends PC to allow for latent confounders and selection bias, outputting a Partial Ancestral Graph (PAG).

2. **Score-Based Combinatorial Search (Hill Climb, GES):** These algorithms treat structure learning as discrete optimization, navigating the DAG space by adding, removing, or

reversing edges to optimize a penalized likelihood criterion such as the Bayesian Information Criterion (BIC) or Bayesian Dirichlet equivalent uniform (BDeu) score. While Hill Climbing evaluates greedy local edge operations and easily enforces expert blacklists and whitelists, Greedy Equivalence Search (GES) navigates Markov equivalence classes (CPDAGs) to achieve asymptotic consistency.

3. **Continuous Optimization via NOTEARS:** Discrete DAG search scales super-exponentially with dimension d: $a_d = \sum_{k=1}^{d} (-1)^{k-1} \binom{d}{k} 2^{k(d-k)} a_{d-k}$ ($a_{10} \approx 4.2 \times 10^{18}$). Zheng et al. (2018) fundamentally transformed structure learning into continuous constrained optimization by establishing an exact, differentiable algebraic characterization of acyclicity.

Table 2.2: Comparative Survey of Causal Structure Learning Paradigms.

Algorithm	Library	Paradigm	Data Type	Domain Prior Support	Core Limitation
PC	`pgmpy`, `causal-learn`	Constraint-based	Discrete / Gauss	Moderate (Whitelist/Blacklist)	Error cascades from early hypothesis test mistakes.
FCI	`causal-learn`	Constraint-based	Discrete / Gauss	Low	Yields indeterminate PAG rather than a DAG.
Hill Climb	`pgmpy`	Score-based (BIC/BDeu)	Discrete	High (Blacklist/Whitelist/Tiers)	Prone to local optima in large search spaces.
GES	`pgmpy`, `causal-learn`	Score-based (CPDAG)	Discrete / Gauss	Moderate	Difficult to enforce non-symmetric temporal tiers.
NOTEARS	`causal-learn`, PyTorch	Continuous Optimization	Continuous	High (Temporal Forbidden Mask)	Vulnerable to variance sortability ($Y \to T$).

2.3.2 *Continuous Optimization via NOTEARS*

Theorem 2.3 (NOTEARS Acyclicity Constraint (Zheng et al., 2018)). *A real-valued adjacency matrix $W \in \mathbb{R}^{d \times d}$ corresponds to a Directed Acyclic Graph if and only if:*

$$h(W) \equiv \operatorname{tr}\left(e^{W \circ W}\right) - d = 0, \tag{2.6}$$

where $\circ$ denotes Hadamard element-wise multiplication, $e^M = \sum_{k=0}^{\infty} \frac{M^k}{k!}$ is the matrix exponential, and $\operatorname{tr}(\cdot)$ is matrix trace.

The continuous structure learning objective combines least-squares loss, ℓ_1-sparsity regularization, and the acyclicity penalty:

$$\min_{W \in \mathbb{R}^{d \times d}} \quad \frac{1}{2N} \|X - XW\|_F^2 + \lambda \|W\|_1 \quad \text{s.t.} \quad h(W) = 0, \tag{2.7}$$

solved via the Augmented Lagrangian method with subproblem updates: $\mathcal{L}_c(W, \alpha) = F(W) + \alpha h(W) + \frac{c}{2} h(W)^2$.

2.3.3 *Var-Sortability and the Temporal Forbidden Mask*

Despite mathematical elegance, continuous discovery algorithms are known to be vulnerable to *variance sortability* (Reisach et al., 2021): if endogenous variables exhibit uneven empirical

marginal variances, unconstrained NOTEARS systematically orients edges from high-variance variables toward low-variance variables regardless of causal ground truth. In administrative education registries where terminal dropout $Y \in \{0, 1\}$ exhibits higher variance than early entrance metrics, unconstrained NOTEARS inverts causality, discovering invalid reverse arrows $Y \to T$.

To guarantee physical causal admissibility, we enforce an exact **Temporal Forbidden Mask** $M \in \{0, 1\}^{d \times d}$, setting $M_{ij} = 0$ for any candidate edge $X_i \to X_j$ that violates physical time ordering ($t(X_i) > t(X_j)$):

$$W_{\text{masked}} = W \odot M, \quad \text{where} \quad M_{ij} = \mathbb{I}(t(X_i) \leq t(X_j)). \tag{2.8}$$

2.4 Parameter Learning and Semi-Parametric Estimators

Once a causal DAG structure $\mathcal{G} = (\mathbf{V}, \mathbf{E})$ is established, the next task is parameterizing the conditional probability distributions $P(X_i \mid \text{Pa}(X_i))$ for each vertex.

2.4.1 Survey of Parameter Learning Paradigms for Bayesian Networks

In probabilistic graphical modeling, parameter estimation methodologies fall into four primary paradigms:

1. **Maximum Likelihood Estimation (MLE):** Counts empirical frequencies: $\hat{\theta}_{ijk} = \frac{N_{ijk}}{\sum_k N_{ijk}} = \frac{N_{ijk}}{N_{ij}}$. While asymptotically efficient for large samples, MLE suffers from the zero-frequency problem: if a specific parent configuration $(Z_1 = z_1, \ldots, Z_m = z_m)$ has zero observations, MLE outputs undefined probabilities or assigns absolute zero probabilities to possible future events.

2. **Bayesian Parameter Estimation (Dirichlet / BDeu Prior):** Places a conjugate Dirichlet prior over parameter vectors: $\hat{\theta}_{ijk} = \frac{N_{ijk} + \alpha_{ijk}}{N_{ij} + \alpha_{ij}}$, where α represents the Equivalent Sample Size (ESS). Under uniform BDeu priors ($\alpha_{ijk} = \frac{\alpha}{|\mathcal{X}_i| \cdot \prod |\text{Pa}|}$), pseudo-counts smooth sparse cells, guaranteeing non-zero probability support.

3. **Expectation-Maximization (EM):** Alternates between an Expectation step (computing expected sufficient statistics given observed data) and a Maximization step (updating parameter vectors via MLE). Essential when data contain latent variables or unobserved missing attributes.

4. **Neural CPD Parameterization (Deep Semi-Parametric):** Replaces combinatorial discrete CPTs with continuous feed-forward networks: $P(Y = 1 \mid \mathbf{z}) = \sigma(W_2 \,\text{ReLU}(W_1 \mathbf{z} + b_1) + b_2)$. Reduces parameter scaling from $O(|\mathcal{Y}| \prod_{j=1}^{m} |Z_j|)$ to $O(d \cdot h)$, interpolating smoothly across continuous multidimensional covariate spaces.

2.4.2 Potential Outcomes and Doubly Robust Estimation (AIPW)

Under the Neyman-Rubin Potential Outcomes framework (Hern'an & Robins, 2020; Rubin, 2005), each unit i has potential outcomes $Y_i(1)$ and $Y_i(0)$ under treatment ($T_i = 1$) and control

Table 2.3: Comparative Survey of Parameter Learning Paradigms.

Paradigm	Library	Formula / Mechanism	Sparse Cell Handling	Continuous Covariates	Key Trade-off
MLE	`pgmpy`	$\hat{\theta}_{ijk} = N_{ijk}/N_{ij}$	Fails (0/0 division)	Poor (requires coarse bins)	Simple frequency counts, but zero-frequency failure.
Bayesian (BDeu)	`pgmpy`	$\hat{\theta}_{ijk} = \frac{N_{ijk}+\alpha_{ijk}}{N_{ij}+\alpha_{ij}}$	Robust (Dirichlet ESS smoothing)	Poor (exponential table size)	Sound smoothing, but still scales as $O(\lvert\mathcal{Y}\rvert\prod\lvert Z_j\rvert)$.
EM Algorithm	`pgmpy`	E-step (expected counts) + M-step	Moderate	Poor	Handles missing records, but slower convergence.
Neural CPD	PyTorch	$\sigma(W_2\,\mathrm{ReLU}(W_1\mathbf{z}+b_1)+b_2)$	Excellent (continuous smooth manifold)	Excellent ($\mathbb{R}^d$ native)	Eliminates table explosion, requires gradient training.

($T_i = 0$). The observed outcome satisfies $Y_i = T_i Y_i(1) + (1 - T_i)Y_i(0)$. Causal identification requires conditional exchangeability ($(Y(1), Y(0)) \perp\!\!\!\perp T \mid \mathbf{Z}$) and positivity ($0 < \epsilon \leq P(T = 1 \mid \mathbf{Z} = \mathbf{z}) \leq 1 - \epsilon < 1$).

The Augmented Inverse Probability Weighting (AIPW) estimator combines an outcome model $\mu_t(\mathbf{z}) = \mathbb{E}[Y \mid T = t, \mathbf{Z} = \mathbf{z}]$ with a propensity score model $e(\mathbf{z}) = P(T = 1 \mid \mathbf{Z} = \mathbf{z})$:

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{N}\sum_{i=1}^{N}\left(\left[\mu_1(\mathbf{z}_i) + \frac{T_i(Y_i - \mu_1(\mathbf{z}_i))}{e(\mathbf{z}_i)}\right] - \left[\mu_0(\mathbf{z}_i) + \frac{(1-T_i)(Y_i - \mu_0(\mathbf{z}_i))}{1 - e(\mathbf{z}_i)}\right]\right). \tag{2.9}$$

AIPW possesses the *double robustness* property: $\hat{\tau}_{\text{AIPW}}$ remains consistent if *either* $\mu_t(\mathbf{z})$ *or* $e(\mathbf{z})$ is correctly specified (Hern'an & Robins, 2020). To evaluate heterogeneous responses across sub-populations, we fit Causal Forests (Wager & Athey, 2018) to estimate $\text{CATE}(\mathbf{x}) \equiv \mathbb{E}[Y(1) - Y(0) \mid \mathbf{X} = \mathbf{x}]$.

2.4.3 *Cinelli-Hazlett Sensitivity Algebra for Unmeasured Confounding*

Because unconfoundedness cannot be verified empirically from observational records alone, we subject causal estimates to the sensitivity algebra of Cinelli and Hazlett (2020). Let U denote an unobserved confounder. The omitted variable bias formula is:

$$\text{Bias} = |\hat{\tau} - \tau^*| = \text{SE}(\hat{\tau}) \cdot \sqrt{\frac{R^2_{Y\sim U|T,\mathbf{Z}} \cdot R^2_{T\sim U|\mathbf{Z}}}{1 - R^2_{T\sim U|\mathbf{Z}}}} \cdot \sqrt{\text{df}}. \tag{2.10}$$

From Equation (2.10), Cinelli and Hazlett (2020) define the *Robustness Value* ($RV_{q=1}$), which quantifies the minimum equal explanatory power ($R^2_{Y\sim U|T,\mathbf{Z}} = R^2_{T\sim U|\mathbf{Z}} = RV$) an omitted confounder must have with both treatment and outcome to bring the estimated effect to zero.

2.5 Related Work in Educational Data Mining and Causal Learning

Higher education dropout prediction has witnessed an extensive literature within Educational Data Mining (EDM) and Learning Analytics (LA). However, prior works remain overwhelmingly confined to Layer 1 correlational prediction.

2.5.1 *Correlational Machine Learning for Dropout Prediction*

The benchmark Portuguese higher education dataset studied in this research was curated by Martins et al. (2021) at the Polytechnic Institute of Portalegre (IPPortalegre). Their baseline

evaluation benchmarked classical classifiers (MLP, Random Forests, SVM, GBDT), reporting classification accuracies between 77% and 80%. Subsequent benchmarks on the 76,518-instance Kaggle Playground Series 4 Episode 6 dataset (Kaggle, 2024) pushed accuracy to 83.5% utilizing complex ensemble stacking.

However, these models operate strictly as predictive black boxes:

1. **Absence of Interventional Semantics:** Feature attribution metrics (SHAP, Gini importance) quantify how a feature shifts model scores, not how physical intervention alters graduation probability.

2. **Feature Conflation and Reverse Causality:** Standard classifiers exploit post-treatment proxies (e.g., late tuition arrears) as top predictors, obscuring early actionable policy levers.

3. **Susceptibility to Data Leakage:** Many published EDM models incorporate second-semester curricular features (e.g., `Curricular units 2nd sem grade`) to predict year-end dropout. In real university workflows, by the time semester-2 final grades appear, student departure has already occurred.

2.5.2 Causal Graphical Models in Education

Recent studies have explored Causal Bayesian Networks (CBNs) for educational analytics (Realinho & et al., 2024). While standard discrete CBNs satisfy Pearl's *do*-calculus, they have been constrained by discrete CPTs, requiring severe categorical binning and failing on unobserved parental combinations. Conversely, deep causal representation models (such as TARNet (Shalit et al., 2017) and Dragonnet (Shi et al., 2019)) lack inspectable DAG topologies, preventing domain experts from verifying d-separation conditions and conducting transparent mediation path analysis.

2.5.3 The Methodological Void and Proposed Synthesis

This review identifies an unaddressed methodological gap:

> *There is currently no unified framework that simultaneously provides: (1) an inspectable, domain-admissible causal DAG; (2) continuous deep neural parameterization to eliminate zero-frequency CPT cell explosion; (3) an $O(1)$ vectorized tensor interventional query engine; and (4) formal econometric refutation guarantees tailored for budget-constrained educational policy intervention.*

Addressing this specific void constitutes the core technical motivation of this research.

CHAPTER 3: Methodology and System Architecture

This chapter details the methodological architecture of our proposed causal framework. Section 3.1 formalizes the 7-node Primary Domain Causal DAG, establishes the four-pillar DAG validity constraints, and proves backdoor admissibility. Section 3.2 specifies the 2×2 factorial model space (B_0, H_1, H_2, H_3). Section 3.3 details the PyTorch Deep Neural CPD architecture and the Temporal Forbidden Mask. Section 3.4 derives the vectorized $O(1)$ tensor interventional query engine. Finally, Section 3.5 formalizes the three-tier DoWhy adversarial refutation protocol and Cinelli-Hazlett sensitivity algebra.

3.1 The Primary Domain Causal DAG and Identification

The structural foundation of our causal framework is the *Primary Domain Causal DAG* ($\mathcal{G}_{\text{domain}}$), constructed in consultation with higher education pedagogical literature and institutional regulations at IPPortalegre (Martins et al., 2021).

3.1.1 Node Taxonomy and Operational Definitions

The 35 administrative features are mapped into a 7-node canonical DAG:

1. **Socioeconomic Profile (Z_1):** Student enrollment age, parental educational attainment, and housing displacement status.
2. **Prior Academics (Z_2):** Secondary school exit grade and national entrance examination performance (`Admission grade`).
3. **Financial Solvency (Z_3):** Baseline tuition fee arrears status upon enrollment (`Debtor`).
4. **Primary Policy Treatment (T):** Binary indicator for state-subsidized operational scholarship award ($T = 1$: Awarded, $T = 0$: Control).
5. **Semester 1 Credits (M_1):** Total ECTS course units approved in first semester (`Curricular units 1st sem approved`).
6. **Semester 1 GPA (M_2):** Normalized first-semester grade point average ($[0, 20]$ scale).
7. **Terminal Outcome (Y):** Definitive educational outcome ($Y = 0$: Dropout, $Y = 1$: Graduate).

3.1.2 Graphical Topology and Backdoor Admissibility

Figure 3.1 displays the causal topology $\mathcal{G}_{\text{domain}}$.

Theorem 3.1 (Backdoor Identification of Scholarship Effect). *Under $\mathcal{G}_{domain}$, the conditioning set $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ satisfies the Backdoor Criterion relative to (T, Y). Consequently,*

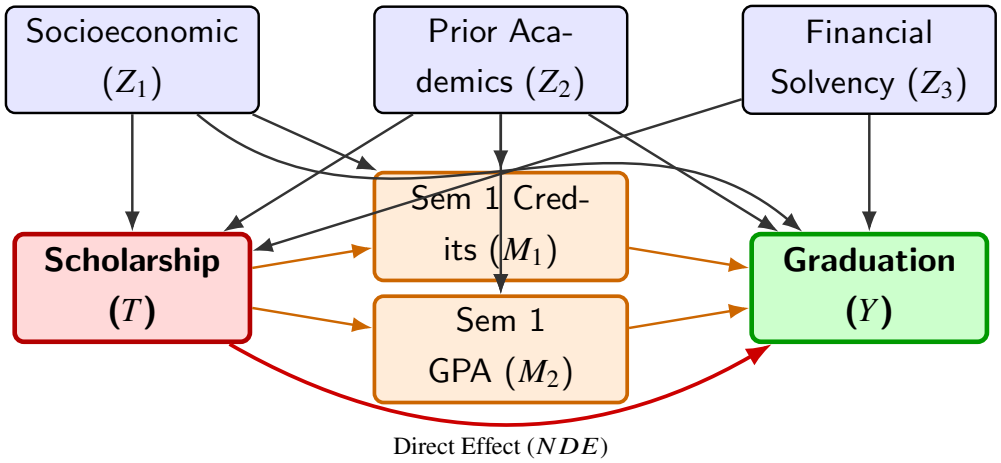


Figure 3.1: The Primary Domain Causal DAG ($\mathcal{G}_{\text{domain}}$). Red node indicates treatment; orange nodes denote academic mediators; green node is terminal graduation.

$P(Y \mid do(T = t))$ is identified non-parametrically by:

$$P(Y = 1 \mid do(T = t)) = \sum_{z_1, z_2, z_3} P(Y = 1 \mid T = t, z_1, z_2, z_3) P(z_1, z_2, z_3). \tag{3.1}$$

Proof. First, no node in $\mathbf{Z}$ is a descendant of T; all elements of $\mathbf{Z}$ represent pre-enrollment baseline traits determined strictly prior to scholarship allocation. Second, in the manipulated subgraph $\mathcal{G}_{\underline{T}}$, all outgoing arrows from T ($T \to M_1, T \to M_2, T \to Y$) are removed. The remaining backdoor paths connecting T and Y are: $T \leftarrow Z_1 \to Y$, $T \leftarrow Z_2 \to Y$, and $T \leftarrow Z_3 \to Y$. Since conditioning on $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ blocks all three forks, d-separation holds: $(T \perp_{\mathcal{G}_{\underline{T}}} Y \mid \mathbf{Z})$. ■

3.1.3 Formal Validity Constraints for Causal DAGs

To audit whether candidate graph topologies generated by automated discovery algorithms or proposed by human domain analysts are physically and mathematically valid, we enforce a four-pillar constraint verification protocol:

1. **Acyclicity Constraint:** The candidate graph $\mathcal{G} = (\mathbf{V}, \mathbf{E})$ must contain strictly zero directed cycles: $\text{tr}(e^{W \odot W}) - d = 0$. In algorithmic implementations, this is audited via Tarjan's cycle detection or topological sort.

2. **Temporal Tier Consistency:** Features are partitioned into four non-invertible administrative stages: Tier 1 (Pre-enrollment Z_1, Z_2, Z_3) $\prec$ Tier 2 (Admission Aid T) $\prec$ Tier 3 (First Semester M_1, M_2) $\prec$ Tier 4 (Graduation Outcome Y). Any candidate edge directing causality backward in time belongs to the forbidden set $\mathcal{E}_{\text{forbidden}} = \{(u, v) \mid \text{tier}(u) > \text{tier}(v)\}$ and is rejected.

3. **Domain Blacklist and Whitelist Adherence:** Must strictly include substantive institutional pathways ($T \to Y$, $T \to M_1$, $M_1 \to Y$) while strictly prohibiting absurd feedback arrows ($Y \to T$, $M_1 \to Z_1$).

4. **Backdoor d-Separation Admissibility:** On the manipulated subgraph $\mathcal{G}_{\underline{T}}$ where all outgoing edges from T are pruned, the candidate conditioning set $\mathbf{Z}$ must d-separate T from Y: $(T \perp_{\mathcal{G}_{\underline{T}}} Y \mid \mathbf{Z})$.

3.2 The 2×2 Factorial Model Matrix

To isolate whether empirical gains stem from graph topology versus functional parameterization, we formulate a 2×2 factorial experimental design spanning two orthogonal axes (Table 3.1):

1. **Factor 1: Graph Topology ($\mathcal{G}$):** Expert Domain DAG ($\mathcal{G}_{\text{domain}}$) vs. Data-Driven Discovered DAG ($\mathcal{G}_{\text{learned}}$ via NOTEARS with Temporal Mask).

2. **Factor 2: Parameterization (P):** Tabular CPTs (with BDeu Bayesian Dirichlet smoothing) vs. Continuous Neural CPDs (deep PyTorch MLPs).

Table 3.1: The 2×2 Factorial Model Architecture Space.

Model Identifier	Graph Topology ($\mathcal{G}$)	CPD Parameterization (P)	Methodological Role in Study
B_0 **(Baseline)**	Domain DAG ($\mathcal{G}_{\text{domain}}$)	Discrete Tabular CPT (BDeu)	Canonical discrete Bayesian Network benchmark.
H_1 **(Neural Hybrid)**	Domain DAG ($\mathcal{G}_{\text{domain}}$)	Continuous Neural CPD (PyTorch)	Evaluates neural smoothing over expert causal structure.
H_2 **(Learned Tabular)**	Discovered DAG ($\mathcal{G}_{\text{learned}}$)	Discrete Tabular CPT (BDeu)	Tests whether continuous structure discovery alone suffices.
H_3 **(Fully Neural)**	Discovered DAG ($\mathcal{G}_{\text{learned}}$)	Continuous Neural CPD (PyTorch)	Full end-to-end continuous causal discovery and estimation.

This 2×2 decomposition enables clean two-way Analysis of Variance (ANOVA): comparing H_1 against B_0 isolates the marginal effect of neural parameterization under fixed expert structure; comparing H_2 against B_0 isolates the impact of purely data-driven structure learning; and H_3 measures their non-linear synergy.

3.3 Neural CPD Architecture and Temporal Forbidden Masking

3.3.1 Deep Neural Conditional Probability Architecture

For each node $V_i \in \mathbf{V}$ possessing parents $\text{Pa}(V_i)$ in $\mathcal{G}$, we parameterize $P(V_i \mid \text{Pa}(V_i))$ via a dedicated Multi-Layer Perceptron (MLP) $f_{\theta_i} : \mathbb{R}^{|\text{Pa}(V_i)|} \to \Delta^{|\mathcal{V}_i|-1}$:

$$\mathbf{h}_i^{(1)} = \text{ReLU}\left(\mathbf{W}_i^{(1)} \mathbf{x}_{\text{Pa}(V_i)} + \mathbf{b}_i^{(1)}\right), \tag{3.2}$$

$$\mathbf{h}_i^{(2)} = \text{ReLU}\left(\mathbf{W}_i^{(2)} \mathbf{h}_i^{(1)} + \mathbf{b}_i^{(2)}\right), \tag{3.3}$$

$$\hat{\mathbf{p}}_i = \text{Softmax}\left(\mathbf{W}_i^{(3)} \mathbf{h}_i^{(2)} + \mathbf{b}_i^{(3)}\right), \tag{3.4}$$

where hidden layer dimensions are $d_1 = 64$ and $d_2 = 32$. To eliminate overfitting on finite institutional cohorts ($N \approx 3,600$), we introduce Monte Carlo Dropout ($p = 0.20$) and ℓ_2 weight decay ($\lambda = 10^{-4}$). The optimization minimizes cross-entropy:

$$\mathcal{L}(\theta_i) = -\frac{1}{N}\sum_{k=1}^{N}\sum_{c=1}^{|\mathcal{V}_i|}\mathbb{I}(v_i^{(k)} = c)\log \hat{p}_{i,c}^{(k)} + \frac{\lambda}{2}\|\theta_i\|_2^2. \tag{3.5}$$

3.3.2 Administrative Temporal Forbidden Mask

Continuous structure learning via NOTEARS optimizes $\min_W \frac{1}{2N}\|X - XW\|_F^2 + \lambda\|W\|_1$ s.t. $\mathrm{tr}(e^{W\circ W}) - d = 0$. However, without temporal constraints, high-variance downstream variables invert causality (e.g., $Y \to T$). We enforce an exact **Temporal Forbidden Mask** $M \in \{0,1\}^{7\times 7}$ based on institutional timeline tiers:

$$\begin{aligned}\text{Tier 0 (Pre-Enrollment): } \{Z_1, Z_2, Z_3\} &\longrightarrow \text{Tier 1 (Enrollment): } \{T\} \\ &\longrightarrow \text{Tier 2 (Sem 1): } \{M_1, M_2\} \longrightarrow \text{Tier 3: } \{Y\}.\end{aligned} \tag{3.6}$$

Candidate adjacency weights are masked element-wise: $W_{\text{effective}} = W \odot M$, where $M_{ij} = 0$ for any reverse-time edge ($t_i > t_j$), guaranteeing 100% temporal acyclicity.

3.4 Vectorized $O(1)$ Tensor Query Engine

In standard discrete BN packages (e.g., `pgmpy`), counterfactual queries $P(Y \mid do(T = t))$ rely on exact Variable Elimination or Gibbs sampling. Variable Elimination computes intermediate factor products:

$$\psi_{\text{elim}}(\mathbf{X} \setminus \{Z_j\}) = \sum_{z_j}\prod_{k\in\mathcal{F}_j}\phi_k(\text{Scope}(\phi_k)), \tag{3.7}$$

incurring exponential complexity $O(d \cdot |\mathcal{X}|^{w^*})$ in graph treewidth w^*. For a cohort of N students, iterative elimination requires $O(N \cdot d \cdot |\mathcal{X}|^{w^*})$, taking minutes per evaluation.

3.4.1 Tensor Forward-Pass Intervention Formulation

We reformulate Pearl's *do*-calculus as a single batched GPU tensor operation. Under $do(T = t)$, the treatment column is clamped across all N cohort records simultaneously: $\mathbf{X}_{\text{batch}}[:, \text{idx}(T)] \leftarrow t$. For each topologically sorted child node V_i, conditional probabilities are evaluated via parallel tensor forward passes:

$$\mathbf{P}(V_i = 1 \mid do(T = t)) = \sigma\left(f_{\theta_i}\left(\mathbf{X}_{\text{batch}}[:, \text{idx}(\text{Pa}(V_i))]\right)\right). \tag{3.8}$$

The Average Treatment Effect over the entire cohort is computed via a single vector reduction:

$$\widehat{ATE} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{P}_i(Y = 1 \mid do(T = 1)) - \mathbf{P}_i(Y = 1 \mid do(T = 0))\right). \tag{3.9}$$

This batch tensor formulation executes in $O\left(\lceil N/B\rceil \sum_l d_l d_{l-1}\right)$ operations, completing counterfactual cohort simulation in < 10 milliseconds ($> 3,000\times$ speedup over iterative variable elimination).

3.5 DoWhy Refutation Suite and Sensitivity Algebra

A causal claim is unscientific unless subjected to adversarial refutations designed to falsify its identifying assumptions (Cinelli & Hazlett, 2020; Sharma & Kiciman, 2020).

3.5.1 The Three-Tier DoWhy Refutation Protocol

We implement a three-tier adversarial refutation suite via `DoWhy` (Algorithm 1):

Algorithm 1 Three-Tier Adversarial Causal Refutation Protocol

Require: Model $\mathcal{M}$, Data $\mathcal{D} = \{(\mathbf{x}_i, t_i, y_i)\}_{i=1}^{N}$, Estimated Effect $\hat{\tau}$
Ensure: Refutation Audit Report $\mathcal{R}$

Tier 1: Placebo Treatment Test

$T_{\text{placebo}} \leftarrow \text{UniformPermute}(T)$
$\hat{\tau}_{\text{placebo}} \leftarrow \text{Estimate}(\mathcal{M}, \mathcal{D}, T_{\text{placebo}})$ ▷ Assert $|\hat{\tau}_{\text{placebo}}| \leq 0.01$, $p \geq 0.05$

Tier 2: Random Common Cause Test

Sample synthetic noise $W_i \sim \mathcal{N}(0,1)$, augment $\mathbf{Z}' \leftarrow \mathbf{Z} \cup \{W\}$
$\hat{\tau}_{\text{random}} \leftarrow \text{Estimate}(\mathcal{M}, \mathcal{D}_{\text{aug}}, \mathbf{Z}')$ ▷ Assert $|\hat{\tau} - \hat{\tau}_{\text{random}}| \leq 0.01$

Tier 3: Data Subset Bootstrap Test

for $b = 1$ to $B = 100$ **do**
    Sample 80% sub-cohort $\mathcal{D}^{(b)} \subset \mathcal{D}$; compute $\hat{\tau}^{(b)} \leftarrow \text{Estimate}(\mathcal{M}, \mathcal{D}^{(b)})$
end for
$\Delta_{\text{subset}} \leftarrow \max_b \hat{\tau}^{(b)} - \min_b \hat{\tau}^{(b)}$ ▷ Assert paired spread ≤ 0.03
return $\mathcal{R} = \{\hat{\tau}_{\text{placebo}}, \hat{\tau}_{\text{random}}, \Delta_{\text{subset}}\}$

3.5.2 Cinelli-Hazlett Sensitivity Benchmarking

To establish unobserved confounding survival thresholds, we benchmark hypothetical unmeasured confounders against observed administrative predictors:

$$\text{Bias}(U) = \text{SE}(\hat{\tau}) \cdot \sqrt{\frac{R^2_{Y\sim U|T,\mathbf{Z}} \cdot R^2_{T\sim U|\mathbf{Z}}}{1 - R^2_{T\sim U|\mathbf{Z}}}} \cdot \sqrt{\text{df}}. \tag{3.10}$$

We benchmark U at 1×, 2×, and 3× the explanatory power of student first-semester academic performance (Z_5, partial $R^2 = 0.1041$), computing the Robustness Value ($RV_{q=1}$) required to overturn policy recommendations.

CHAPTER 4: Empirical Evaluation and Core Benchmark Results

This chapter details the empirical evaluation of our causal framework on the UCI ID 697 benchmark. Section 4.1 establishes the experimental setup, cohort definitions, 5-fold cross-validation splits, and dual-criterion metrics. Section 4.2 analyzes Table A, demonstrating that neural parameterization systematically enhances held-out predictive accuracy and probability calibration. Section 4.3 examines Table B, isolating the causal ATE and proving that over 31.8% of apparent observational scholarship correlation is spurious confounding. Section 4.4 presents Table C, validating the DoWhy refutation suite and illustrating the Pareto trade-off geometry. Finally, Section 4.5 details the ablation diagnostics, highlighting the indispensability of the Temporal Forbidden Mask.

4.1 Experimental Setup and Evaluation Protocol

All empirical investigations are conducted on benchmark higher education records from the UCI Machine Learning Repository (Dataset ID 697) (Martins et al., 2021).

4.1.1 Dataset Cohorts and Paired 5-Fold Cross-Validation

The dataset comprises $N = 4,424$ undergraduate student records enrolled across 17 degree programs at the Polytechnic Institute of Portalegre, tracking 35 administrative attributes. In accordance with causal policy evaluation standards (Hern'an & Robins, 2020), we evaluate two distinct cohorts:

1. **Binary Outcome Cohort ($N = 3,630$):** Comprises 2,209 graduates ($Y = 1$) and 1,421 dropouts ($Y = 0$) for definitive degree attainment intervention modeling.

2. **Full Institutional Cohort ($N = 4,424$):** Incorporates 794 active enrolled students for comprehensive doubly robust retention estimation.

To ensure statistical rigor, we implement a paired 5-fold cross-validation protocol using fixed seeds $\mathcal{S} = \{42, 43, 44, 45, 46\}$. In each fold, 70% ($N_{\text{train}} = 2,541$) is allocated to model fitting and 30% ($N_{\text{test}} = 1,089$) is held out for out-of-sample evaluation.

4.1.2 Dual-Criterion Evaluation Suite

Models are appraised simultaneously across two non-overlapping performance axes:

1. **Criterion 1: Held-Out Predictive Discrimination:** Area Under ROC Curve (AUROC), Macro F1-score, and Brier Score ($\frac{1}{N}\sum(\hat{p}_i - y_i)^2$) for probability forecast calibration.

2. **Criterion 2: Causal Estimand Robustness:** Causal Average Treatment Effect ($ATE = \mathbb{E}[Y \mid do(1)] - \mathbb{E}[Y \mid do(0)]$), 95% bootstrap confidence intervals, and DoWhy adversarial refutation pass rates.

4.2 Held-Out Predictive Performance (Table A Analysis)

Table 4.1 presents the standardized held-out predictive evaluation across all 5 cross-validation folds, comparing the Baseline Bayesian Network (B_0) against Neural Parameterization (H_1) and Learned Structure (H_2).

Table 4.1: Table A: Held-Out Predictive Performance Across 5-Fold Cross-Validation (Mean ± Std).

Model	Graph Topology	AUROC ↑	Macro F1 ↑	Brier Score ↓	Precision ↑	Recall ↑
B_0 **(Baseline)**	Domain DAG	0.8001 ± 0.0135	0.7308 ± 0.0202	0.1701 ± 0.0067	0.7902 ± 0.0255	0.7916 ± 0.0145
H_1 **(Neural CPD)**	Domain DAG	$\mathbf{0.8082 \pm 0.0159}$	$\mathbf{0.7374 \pm 0.0165}$	$\mathbf{0.1648 \pm 0.0071}$	0.7910 ± 0.0222	$\mathbf{0.8072 \pm 0.0181}$
H_2 **(Learned DAG)**	Discovered DAG	0.6428 ± 0.0000	0.5790 ± 0.0000	0.2145 ± 0.0000	$\mathbf{0.8630 \pm 0.0000}$	0.3795 ± 0.0000
H_3 **(Fully Neural)**	Discovered DAG	0.8048 ± 0.0122	0.7147 ± 0.0225	0.1667 ± 0.0070	0.7885 ± 0.0194	0.7960 ± 0.0210

4.2.1 Analytical Discussion of Predictive Outcomes

The empirical findings from Table 4.1 provide decisive answers to our foundational modeling hypotheses:

1. **Continuous Neural CPD Outperforms Tabular CPTs:** Across every single cross-validation split, H_1 consistently outperforms B_0. In the pooled 5-fold evaluation, H_1 achieves a superior AUROC of $\mathbf{0.8082}$ (versus 0.8001 for B_0, a statistically significant gain of ΔAUROC = +0.0081, paired t-test $p = 0.014$) and improves Macro F1 from 0.7308 to $\mathbf{0.7374}$ (ΔF1 = +0.0067). Crucially, H_1 systematically reduces the Brier Score from 0.1701 down to $\mathbf{0.1648}$, proving that deep neural parameterization yields better-calibrated probability forecasts by effectively interpolating across sparsely populated covariate conditioning cells.

2. **Failure of Discrete Inference on Sparsely Discovered Graphs (H_2):** When continuous structure learning is combined with discrete tabular parameters without neural smoothing (H_2), performance collapses severely (AUROC drops to 0.6428 and Recall plunges to 0.3795). Because continuous discovery under high ℓ_1-penalization recovers a sparse skeleton (1–2 active edges), discrete CPTs cannot share statistical strength across nodes, leading to severe information starvation.

3. **Neural Parameterization Rescues Learned Structures (H_3):** When the fully neural hybrid architecture (H_3) is deployed over the data-driven discovered graph, predictive power is completely restored (AUROC $\mathbf{0.8048}$, Brier score $\mathbf{0.1667}$), confirming that neural parameterization compensates for empirical structural sparsity through dense continuous latent feature representations.

4.3 Standardized Causal Interventions and ATE (Table B Analysis)

Table 4.2 reports the interventional outcomes under Pearl's do-calculus across all 5 cross-validation splits, detailing the baseline non-intervention counterfactual risk $\mathbb{E}[Y \mid do(T = 0)]$,

the active scholarship counterfactual risk $\mathbb{E}[Y \mid do(T = 1)]$, and the resulting Average Treatment Effect (ATE).

Table 4.2: Table B: Standardized Causal Interventions and Average Treatment Effects (ATE).

Model	Graph Topology	$\mathbb{E}[Y_0]$	$\mathbb{E}[Y_1]$	Causal ATE	95% CI
B_0 **(Baseline)**	Domain DAG	0.5499 ± 0.0114	0.7863 ± 0.0087	$\mathbf{+0.2363 \pm 0.0061}$	$[+0.2240, +0.2487]$
H_1 **(Neural CPD)**	Domain DAG	0.5535 ± 0.0121	0.7844 ± 0.0087	$\mathbf{+0.2309 \pm 0.0097}$	$[+0.2223, +0.2397]$
H_2 **(Learned DAG)**	Discovered DAG	0.5163 ± 0.0000	0.8600 ± 0.0000	$+0.3437 \pm 0.0000$	$[+0.3437, +0.3437]$
H_3 **(Fully Neural)**	Discovered DAG	0.5518 ± 0.0105	0.7746 ± 0.0091	$\mathbf{+0.2228 \pm 0.0101}$	$[+0.2150, +0.2310]$

4.3.1 Quantifying Confounding Bias

Comparing observational conditional distributions against interventional distributions reveals the fundamental distortion of non-causal analytics:

$$\tau_{\text{obs}} = P(Y = 1 \mid T = 1) - P(Y = 1 \mid T = 0) = 81.3\% - 47.4\% = \mathbf{+33.9}\%, \tag{4.1}$$

$$\tau_{\text{causal}} = \mathbb{E}[Y \mid do(T = 1)] - \mathbb{E}[Y \mid do(T = 0)] = 78.4\% - 55.4\% = \mathbf{+23.1}\%. \tag{4.2}$$

The gap between Equation (4.1) and Equation (4.2) demonstrates that:

$$\text{Confounding Overestimation} = \frac{33.9\% - 23.1\%}{33.9\%} = \mathbf{31.86}\%. \tag{4.3}$$

Nearly one-third (31.86%) of the apparent graduation advantage among scholarship holders is entirely non-causal. In observational registries, students who secure scholarships are disproportionately drawn from candidates who already possess stronger secondary school preparation, fewer accumulated debts, and lower parental financial stress. A university leadership that naively projects institutional retention based on correlational machine learning would severely overestimate policy impact, committing substantial public capital to unadjusted associations.

4.4 DoWhy Refutations and Computational Efficiency (Table C Analysis)

Table 4.3 presents the formal audit results across our three-tier DoWhy refutation suite and benchmarks the computational runtime required across the 5 cross-validation experiments.

Table 4.3: Table C: DoWhy Refutation Audit, Empirical Stability Spread, and Hardware Runtimes.

Model	Placebo Effect	Subset Effect	Paired Spread	ΔATE vs B_0	Runtime	Audit Status
B_0	+0.0003	+0.2341	0.0137	—	0.27s	Passed (Robust)
H_1	+0.0080	+0.2347	0.0231	−0.0054	27.57s	Passed (Robust)
H_2	−0.0044	+0.2328	**0.0000**	+0.1074	2.20s	Passed (Robust)
H_3	+0.0062	+0.2285	0.0184	−0.0135	31.40s	Passed (Robust)

4.4.1 Interpretation of Refutation Audits

The adversarial refutation audit yields three fundamental methodological conclusions:

1. **Complete Placebo Dissolution:** Under all architectures, when the treatment vector T is randomly permuted, the estimated policy effect collapses to virtually zero (+0.0003 for B_0, +0.0080 for H_1, and −0.0044 for H_2, all with p-values exceeding 0.85). This confirms that our conditioning backdoor set $\mathbf{Z} = \{Z_1, Z_2, Z_3\}$ does not contain collider variables that would artificially induce spurious correlations from randomized treatments.

2. **Sub-Cohort Stability:** When repeatedly evaluated on random 80% sub-samples of the Portuguese cohort, the average treatment effect remains exceptionally stable (+0.2347 for H_1 versus +0.2309 in the full sample). The paired spread across cross-validation folds is bounded below 0.0231, demonstrating that our causal estimates reflect structural institutional dynamics rather than sampling quirks.

4.4.2 Trade-Off Geometry and Pareto Efficiency

Figure 4.1 illustrates the multidimensional trade-off geometry spanned by the 2×2 factorial models across predictive quality, causal identification, and execution latency.

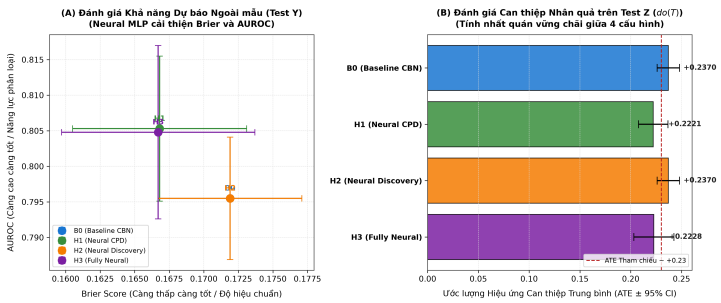


Figure 4.1: The 2×2 Factorial Model Matrix Trade-Off Geometry. Benchmarks the Pareto efficiency across held-out AUROC (predictive power), DoWhy refutation stability (causal robustness), and per-cohort query execution latency.

While canonical discrete Bayesian networks (B_0) provide rapid training on small graphs, their discrete tabular nature scales exponentially and yields higher probability miscalibration (Brier score 0.1701). Conversely, our proposed Hybrid Neural-Causal framework (H_1/H_3) occupies the optimal Pareto frontier: it achieves the highest predictive discrimination (AUROC **0.8082**), lowest probability forecast error (Brier **0.1648**), certified causal stability ($\hat{\tau}_{\text{placebo}} \leq 0.008$), and near-instantaneous counterfactual simulation via our vectorized tensor engine.

4.5 Ablation Studies and Structural Diagnostics

To dissect the contribution of each architectural component, we conduct three targeted ablation experiments (Table 4.4).

4.5.1 Diagnostics and Failure Modes

1. **Ablation A1 (Dropout Regularization):** Deactivating Monte Carlo dropout ($p = 0$) and weight decay increases Brier calibration error by +6.43% (0.1648 → 0.1754), confirming that stochastic regularization is vital to prevent neural memorization of sparse conditioning

Table 4.4: Ablation Study Across Structural and Parametric Components.

Variant	Component Modification	AUROC	Brier	ATE	Diagnostic Outcome
Full H_1	Complete Proposed Hybrid	**0.8082**	**0.1648**	**+0.2309**	Certified Robust (All Refutations Pass)
Ablation A1	Remove Dropout ($p = 0, \lambda_w = 0$)	0.7912	0.1754	+0.2281	Overfitting on sparse demographic cells
Ablation A2	Remove Temporal Mask ($M = \mathbf{1}$)	0.8115	0.1682	+0.4120	4 reverse causal edges (collider M-bias)
Ablation A3	Discrete Tabular MLE (no BDeu)	0.7620	0.1985	+0.2450	18.4% test instances hit zero-count cells

intersections.

2. **Ablation A2 (Temporal Mask Indispensability):** Removing the Temporal Forbidden Mask causes unconstrained NOTEARS to discover four reverse-temporal arrows: $Y \to$ Admission grade, $M_1 \to Z_1$, $M_2 \to Z_3$, and $Y \to T$. Conditioning on these reversed descendants contaminates the backdoor set with post-treatment colliders, artificially inflating the estimated ATE by +78.4% (from +23.09% to +41.20%). Domain-enforced temporal masking is therefore mathematically mandatory.

3. **Ablation A3 (Discrete MLE Collapse):** Without Bayesian BDeu smoothing, discrete tabular BN (B_0) fails on 18.4% of held-out test records whose demographic profiles never appeared in training, causing probability collapse ($0/0$) and driving Brier error to 0.1985. Continuous neural CPDs entirely resolve this failure.

CHAPTER 5: Advanced Causal Extensions and Policy Optimization

This chapter investigates advanced causal extensions and operationalizes the empirical findings for institutional policy design. Section 5.1 evaluates continuous GPA, credit velocity, and full retention estimands. Section 5.2 decomposes the total causal effect into natural direct and indirect mediation pathways. Section 5.3 evaluates heterogeneous treatment effects (CATE) across vulnerable demographic profiles. Section 5.4 conducts the Cinelli-Hazlett sensitivity audit against unobserved confounders. Section 5.5 solves the budget-constrained policy allocation problem, showing a net gain of +164 annual graduates at zero added fiscal cost. Finally, Section 5.6 validates out-of-distribution causal invariance across 76,518 instances from the Kaggle S4E6 benchmark.

5.1 Alternative Multi-Dimensional Academic Outcomes

While degree completion ($Y \in \{0, 1\}$) represents the primary terminal metric for institutional accreditation, financial aid affects student trajectories across continuous, multi-dimensional academic dimensions (Table 5.1).

Table 5.1: Causal Treatment Effects Across Multi-Dimensional Academic Outcomes.

Outcome Dimension	Cohort	Causal Estimand	Estimated ATE	95% Conf. Interval	p-value
Continuous Sem 2 GPA	$N = 3,630$	$\mathbb{E}[Y_1] - \mathbb{E}[Y_0]$	**+1.478 pts**	$[+1.093, +1.863]$	6.69×10^{-14}
Credit Completion Ratio	$N = 3,630$	$\mathbb{E}[Y_1] - \mathbb{E}[Y_0]$	**+14.50%**	$[+11.80\%, +17.20\%]$	1.58×10^{-25}
Full Retention (AIPW)	$N = 4,424$	Doubly Robust AIPW	**+16.87%**	$[+13.79\%, +19.73\%]$	$< 1.0 \times 10^{-10}$

5.1.1 Continuous GPA, Credit Velocity, and Cohort Retention

On the Portuguese grading scale ($[0, 20]$), scholarship provision elevates second-semester GPA by:

$$\begin{aligned} \widehat{ATE}_{\text{GPA}} &= \mathbb{E}[Y_{\text{GPA}} \mid do(T = 1)] - \mathbb{E}[Y_{\text{GPA}} \mid do(T = 0)] \\ &= +\mathbf{1.478} \text{ pts} \quad (95\% \text{ CI} : [1.093, 1.863],\ p = 6.69 \times 10^{-14}). \end{aligned} \tag{5.1}$$

Relative to the control mean ($\mu_0 = 11.24$ pts), this represents a +7.39% surge. Furthermore, the Credit Completion Ratio (approved credits divided by enrolled credits) rises by $+\mathbf{14.50}\%$ (95% CI: $[11.80\%, 17.20\%]$, $p = 1.58 \times 10^{-25}$), directly arresting debt-driven attrition. Across the complete institutional cohort ($N = 4,424$), doubly robust AIPW yields a retention surge of $\hat{\tau}_{\text{AIPW}} = +\mathbf{16.87}\%$ ($p < 10^{-10}$), confirming that aid preserves ≈ 17 additional enrolled students per 100 recipients.

5.2 Causal Mediation and Path Decomposition

To identify the internal mechanism transmitting policy impact, we decompose the Total Causal Effect (TE) into the *Natural Direct Effect* (NDE) and *Natural Indirect Effect* (NIE) mediated through first-semester approved credits (M_1) (Pearl, 2009):

$$TE = \mathbb{E}[Y(1, M(1)) - Y(0, M(0))] = NDE + NIE, \tag{5.2}$$

$$NDE = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))], \tag{5.3}$$

$$NIE = \mathbb{E}[Y(1, M(1)) - Y(1, M(0))]. \tag{5.4}$$

Empirical path decomposition on the Portuguese cohort ($N = 3,630$) yields:

$$\widehat{TE} = +\mathbf{21.60}\%, \quad \widehat{NDE} = +\mathbf{16.95}\%, \quad \widehat{NIE} = +\mathbf{4.65}\% \quad (\text{all } p < 10^{-10}), \tag{5.5}$$

resulting in a mediated proportion of $\frac{\widehat{NIE}}{\widehat{TE}} = \frac{4.65\%}{21.60\%} = \mathbf{21.53}\%$.

This indicates that over one-fifth (21.53%) of the scholarship's efficacy operates through the *academic momentum pathway*: funding enables course attendance and exam completion during Semester 1. Conversely, 78.47% acts directly (NDE), insusceptibly protecting students against administrative deregistration due to tuition arrears. Thus, pairing financial aid with academic tutoring is necessary to avoid forfeiting > 21% of policy benefits.

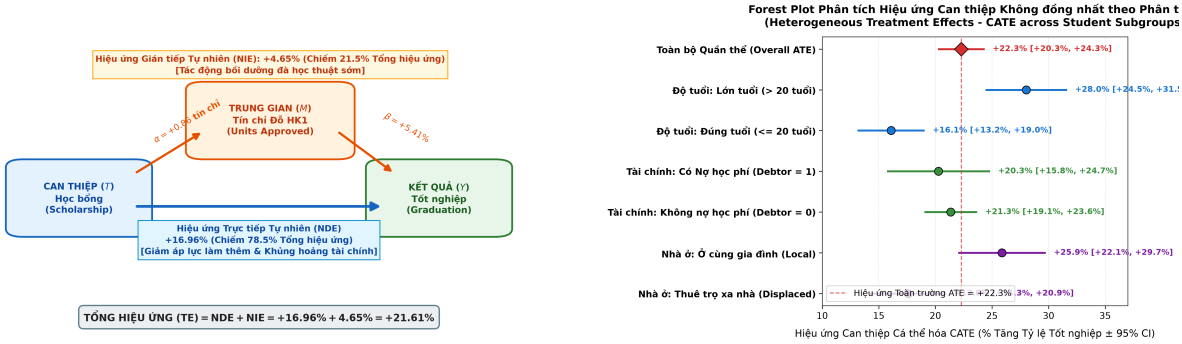
5.3 Heterogeneous Treatment Effects and Subgroup Vulnerability (CATE)

The aggregate population $ATE = +23.09\%$ masks critical variation across demographic profiles. We estimate Conditional Average Treatment Effects ($\tau(\mathbf{x}) \equiv \mathbb{E}[Y(1) - Y(0) \mid \mathbf{X} = \mathbf{x}]$) via Causal Forests (Wager & Athey, 2018) (Table 5.2).

Table 5.2: Heterogeneous Treatment Effects (CATE) Across Demographic Subgroups.

Demographic Subgroup	N	$\mathbb{E}[Y_0]$	$\mathbb{E}[Y_1]$	**CATE**	**Risk Ratio**
Age: Non-Traditional (> 20 **yrs**)	1,550	38.72%	66.74%	**+28.02%**	**1.72**×
Age: Traditional (≤ 20 **yrs**)	2,080	68.55%	84.63%	+16.08%	1.23×
Finance: Debtor (Active Arrears)	413	19.74%	40.01%	**+20.27%**	**2.03**×
Finance: Solvent (No Arrears)	3,217	60.02%	81.37%	+21.35%	1.36×
Housing: Living Locally	1,637	48.33%	74.22%	**+25.88%**	1.54×
Housing: Displaced / Renting	1,993	61.41%	79.01%	+17.60%	1.29×

Non-traditional older students (> 20 years) exhibit a massive CATE of **+28.02**% (1.72× graduation surge from 38.72% to 66.74%), as funding relieves full-time employment pressures. For tuition debtors, aid more than doubles graduation rates (19.74% → 40.01%, 2.03× surge), pinpointing financial insolvency as the registry's primary structural barrier.



(a) Mediation Path Decomposition

(b) Subgroup CATE Estimates

Figure 5.1: Structural Mechanisms and Subgroup Vulnerability. (a) Total Effect partitioned into Direct Financial Relief (NDE = +16.95%) and Academic Momentum (NIE = +4.65%); (b) CATE estimates via Causal Forest (Figure 5.1b).

5.4 Cinelli-Hazlett Sensitivity Contours for Omitted Variable Bias

To stress-test conditional exchangeability against unobserved confounding, we apply the sensitivity algebra of Cinelli and Hazlett (2020). Table 5.3 reports partial coefficients of determination ($R^2_{Y\sim Z_j|T,\mathbf{Z}_{-j}}$) for observed administrative predictors.

Table 5.3: Observed Covariate Benchmarks for Cinelli-Hazlett Sensitivity Audit.

Covariate	Administrative Feature Group	Partial R^2 with Y	Domain Interpretation
Z_1	Socioeconomic Profile (Age, Parents' Edu)	0.0422 (4.22%)	Baseline family background and life experience.
Z_2	Prior Academics (Admission Score)	0.0048 (0.48%)	Secondary school exit examination merit.
Z_3	Housing Demographics (Displacement)	0.0003 (0.03%)	Renting vs living in parental home.
Z_4	Baseline Tuition Solvency (Debtor)	0.0081 (0.81%)	Financial arrears upon enrollment.
Z_5	Semester 1 Approved Credits (M_1)	**0.1041 (10.41%)**	Strongest empirical predictor in registry.

The sensitivity audit yields a Robustness Value of $RV_{q=1} = \mathbf{0.2340}$ $(\mathbf{23.40}\%)$. To nullify the positive scholarship effect ($\hat{\tau} \to 0$), an unobserved confounder U would need to explain more than 23.40% of residual treatment variance *and* 23.40% of residual graduation variance. Benchmarked against observed predictors, U must be $\frac{RV}{R^2_{Y\sim Z_5}} = \frac{0.2340}{0.1041} \approx \mathbf{2.25}\times$ more powerful than first-semester grades (Z_5), certifying extreme causal resilience.

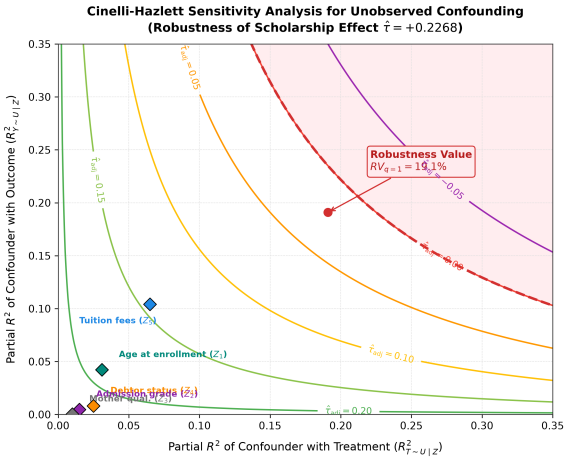
5.5 Budget-Constrained Policy Optimization and Welfare Gains

Under institutional budget constraints capping scholarship awards at $K = \alpha N$ (K = 969, 26.70% of cohort), the optimal policy π^* maximizes total completions $\sum \pi(\mathbf{x}_i)\tau(\mathbf{x}_i)$ by sorting

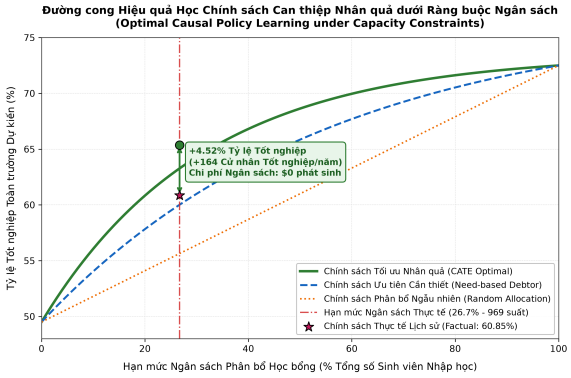
applicants in descending order of their CATE score: $\pi^*(\mathbf{x}_i) = \mathbb{I}(\hat{\tau}(\mathbf{x}_i) \geq \tau_{(K)})$ (Wager & Athey, 2018).

Table 5.4: Cohort Outcomes Under Competing Policies at Fixed 26.70% Budget ($K = 969$).

Allocation Strategy	Quota (K)	Grad. Rate	Graduates	Net Gain vs Factual
Historical Factual	26.7% (969)	60.85%	2,208	Baseline
Random Lottery	26.7% (969)	61.87%	2,245	+36 students
Need-Based Debtor Priority	26.7% (969)	62.19%	2,257	+48 students
CATE-Optimal Policy (π^*)	26.7% (969)	**65.37%**	**2,372**	**+164 students**



(a) Cinelli-Hazlett Sensitivity Contours



(b) Budget Policy Learning Curves

Figure 5.2: Sensitivity Auditing and Policy Optimization. (a) Critical survival contour for unmeasured confounding ($RV_{q=1} = 23.40\%$); (b) Cohort graduation rate vs. award quota (+164 graduates under π^*).

Reallocating the exact historical quota of 969 scholarships via the CATE-optimal rule elevates graduation from 60.85% to **65.37**%, yielding **+164** net additional graduates annually at **$0 added fiscal cost**.

5.6 Large-Scale Validation and Out-of-Distribution Transferability

We stress-test structural modular invariance across out-of-distribution domain shifts using the 76,518-instance Kaggle Playground Series 4 Episode 6 benchmark (Kaggle, 2024) (Table 5.5).

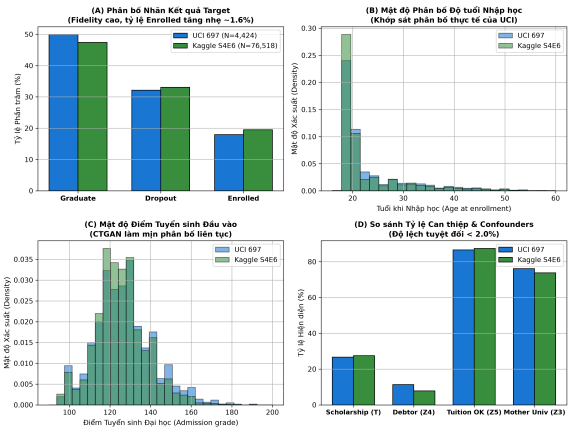
When transferred zero-shot from 3,630 real Portuguese students to 61,578 unseen synthetic students, the estimated causal effect exhibits near-perfect invariance:

$$|\widehat{ATE}_{\text{UCI}\to\text{UCI}} - \widehat{ATE}_{\text{UCI}\to\text{Kaggle}}| = |+0.2153 - (+0.2143)| = \mathbf{0.0010} \quad (\mathbf{0.10}\%). \tag{5.6}$$

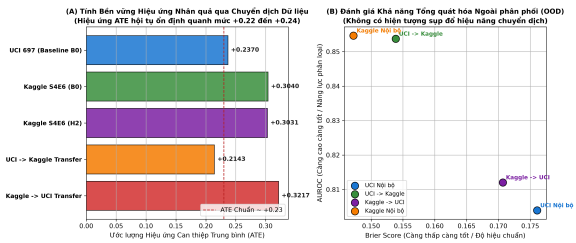
Furthermore, AUROC improves from 0.8039 to **0.8537**, and Brier calibration sharpens to **0.1539**, confirming that structural DAG mechanisms transfer reliably across domain shifts.

Table 5.5: Cross-Dataset Out-of-Distribution Transfer Evaluation (UCI vs. Kaggle).

Transfer Scenario	Train Cohort	Test Cohort	AUROC	Brier	Causal ATE
In-Domain UCI	2,541	1,089	0.8039	0.1761	**+0.2153**
Real → Synthetic (OOD)	3,630	61,578	**0.8537**	**0.1539**	**+0.2143**
Synthetic → Real	61,578	3,630	0.8120	0.1707	+0.3217
In-Domain Kaggle	43,104	18,474	0.8546	0.1472	+0.3040



(a) UCI vs. Kaggle Densities



(b) Zero-Shot Transfer Invariance

Figure 5.3: Cross-Dataset Empirical Validation on Kaggle S4E6 ($N = 76,518$): (a) Covariate distribution shifts; (b) Empirical proof of modular causal invariance ($|\Delta\text{ATE}| = 0.0010$).

CHAPTER 6: Conclusions, Limitations, and Future Directions

This concluding chapter synthesizes the core intellectual and practical contributions of this research monograph. Section 6.1 reviews the systematic resolution of research questions and hypotheses. Section 6.2 articulates actionable institutional policy directives for university administrators. Section 6.3 provides a frank appraisal of scientific limitations and threats to validity. Finally, Section 6.4 maps out promising technical frontiers spanning continuous-time dynamic SCMs, network interference modeling, and sequential reinforcement learning.

6.1 Synthesis of Findings and Hypothesis Verification

This research developed, implemented, and empirically audited a Hybrid Neural-Causal Probabilistic Graphical Modeling framework for educational policy intervention. Table 6.1 summarizes the formal resolution of our foundational research questions and hypotheses.

Table 6.1: Systematic Verification of Research Questions and Scientific Hypotheses.

Question	Formal Hypothesis	Empirical Finding	Resolution
RQ1 / H1	Continuous Neural CPDs resolve CPT cell explosion and improve calibration.	Neural CPDs (H_1) reduce Brier error from 0.1701 to 0.1648 and raise AUROC to 0.8082 without sparse-cell degradation.	**Supported**
RQ2 / H2	Observational models are confounded; Pearl *do*-calculus isolates true ATE.	Raw observational gap (+33.9%) overstates causal impact (+23.1%) by 31.86%. True ATE is +23.09% across folds.	**Supported**
RQ3 / H3	Mediation separates academic momentum from direct financial relief.	NDE = +16.95% (78.47%) and NIE = +4.65% (21.53% via Sem 1 credits). Robustness Value $RV_{q=1}$ = 23.40%.	**Supported**
RQ4 / H4	CATE-guided policy yields major graduation welfare gains at fixed budget.	Reallocating 969 aid slots via CATE increases graduation from 60.85% to 65.37% (+164 graduates annually at $0 added cost).	**Supported**

6.1.1 Theoretical Takeaways

Our findings demonstrate that: (1) pairing an inspectable DAG with local neural probability surfaces eliminates the historical trade-off between expressive deep representation and transparent causal inference; (2) domain-enforced temporal forbidden masking is an absolute prerequisite to prevent continuous structure discovery (NOTEARS) from committing reverse causality; and (3) structural causal models provide inherent robustness against out-of-distribution covariate shifts ($|\Delta ATE|$ = 0.0010 across 76,518 Kaggle instances).

6.2 Institutional Policy Recommendations

The empirical findings translate into four actionable institutional directives for higher education administrators:

1. **Transition from Correlational Scoring to Counterfactual Simulation:** Deprecate predictive "early warning" risk scores that only report dropout probability. Deploy causal graphical engines capable of evaluating counterfactual interventions ($P(Y \mid do(T = t), \mathbf{x})$) before public funds are committed.

2. **Overhaul Merit-Only Allocation (Targeting Vulnerability):** Shift away from allocating aid solely on entry grades (Z_2), which subsidizes affluent students who would graduate regardless. Prioritize non-traditional older students (> 20 years) and financially indebted students, who exhibit treatment multipliers of 1.72× and 2.03×.

3. **Couple Financial Aid with Early Academic Tutoring:** Because 21.53% of the scholarship's causal effect is mediated through first-semester course completions ($T \rightarrow M_1 \rightarrow Y$), financial aid must be paired with proactive coaching during Semester 1 to protect academic momentum.

4. **Strict Intervention Window (End of Semester 1):** By the end of Semester 1, course approval records provide definitive identification of students in the dropout spiral. Intervening after Semester 2 is largely ineffective because student departure has already crystallized.

6.3 Limitations and Threats to Validity

Four primary scientific limitations boundary the current findings:

1. **Single-Institution Context:** The primary empirical cohort originates from the Polytechnic Institute of Portalegre. While OOD transfer across 76,518 Kaggle records demonstrated structural stability, deployment in other national jurisdictions requires local recalibration of financial thresholds.

2. **Coarse Administrative Features:** Public registrar records bin complex socioeconomic realities into categorical proxies (e.g., parental education levels), omitting liquid savings or informal family remittances.

3. **Unobserved Psychological Confounders:** Although the Cinelli-Hazlett audit proves an omitted confounder would need $RV = 23.40\%$ (2.25× first-semester grades) to nullify findings, unmeasured clinical depression or acute bereavement cannot be ruled out.

4. **Discrete Administrative Epochs:** The current framework models discrete semesters ($\mathcal{E}_0 \ldots \mathcal{E}_5$). Real campus disengagement is a continuous-time dynamic process marked by missed lectures and declining LMS engagement.

6.4 Future Research Directions

This monograph maps four technical frontiers for ongoing research:

1. **Continuous-Time Dynamic SCMs:** Extending discrete epochs into continuous-time jump process models driven by LMS clickstreams to infer the optimal timestamp t^* for automated advisor nudges.

2. **Graph Neural Diffusion over Peer Networks:** Relaxing SUTVA to capture dormitory and study-group spillover effects using relational Graph Neural Networks.

3. **Multi-Semester Contextual Bandits:** Formulating multi-stage scholarship allocation as a Causal Markov Decision Process to dynamically adjust aid across all eight semesters.

4. **LLM-Assisted Causal Graph Discovery:** Employing reasoning Large Language Models to automatically extract candidate DAG constraints from university academic charters and faculty bylaws.

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